

名状しがたいお茶会

ハッシュタグ
#teaparty2015

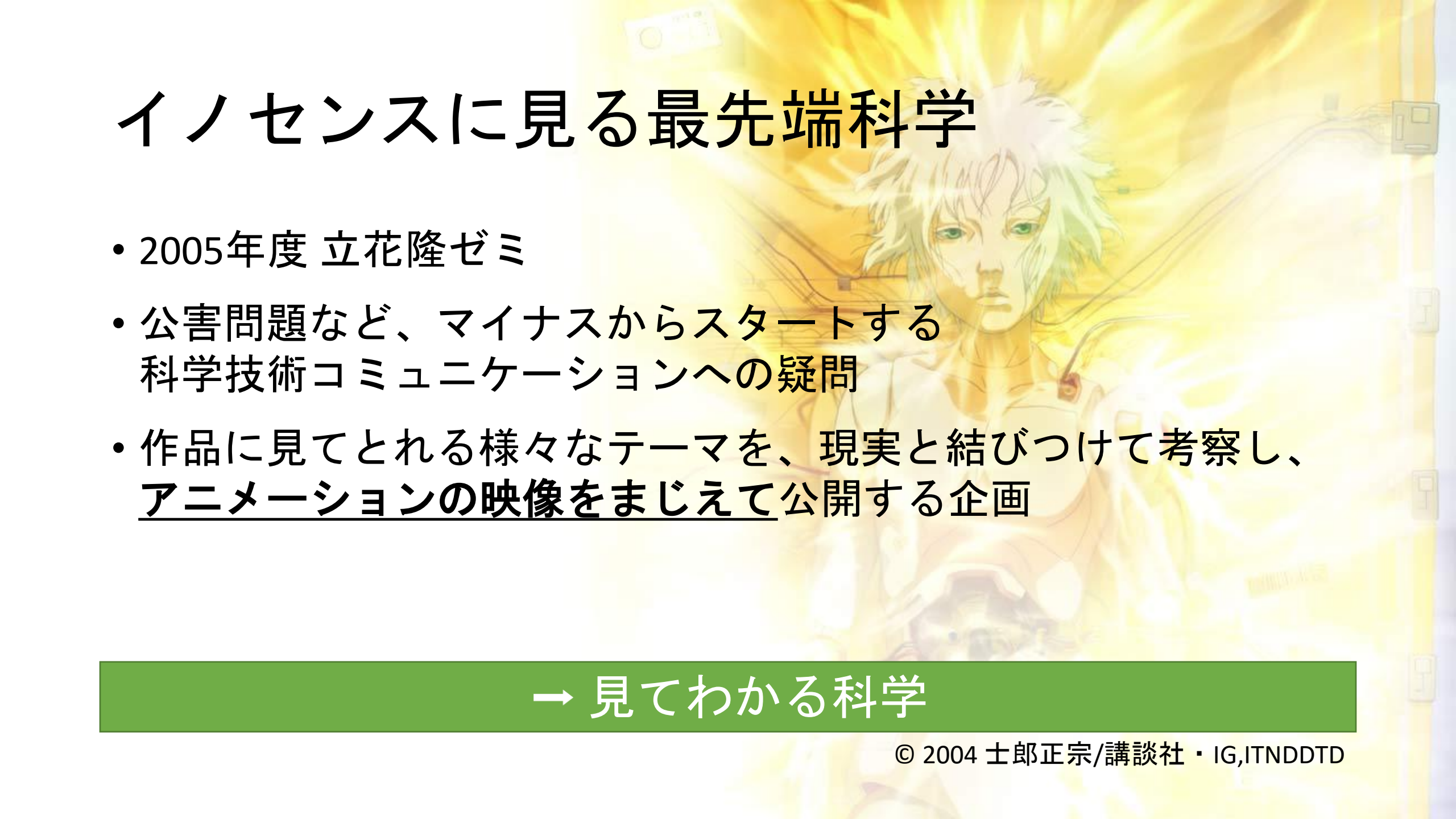
百考は一行にしかず
―これからの科学論―

産業技術総合研究所

加藤 淳

(@arcatdmz, http://junkato.jp)

イノセンスに見る最先端科学



- 2005年度 立花隆ゼミ
- 公害問題など、マイナスからスタートする
  科学技術コミュニケーションへの疑問
- 作品に見てとれる様々なテーマを、現実と結びつけて考察し、
  アニメーションの映像をまじえて公開する企画

→ 見てわかる科学

© 2004 士郎正宗/講談社・IG,ITNDDTD

科学論文

- 科学の基本
- 図表が**意外と**重要

“Orality and Literacy” [Ong ‘82]

文字表現

抽象的ロジックの精密な表現が得意

画像表現

具体例を分かりやすく表現できる

[Kato, et al., 2013]

Picode:
Inline Photos Representing Posture Data in Source Code

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ABSTRACT
Current programming environments use textual or symbolic representations. While these representations are appropriate for describing logical processes, they are not appropriate for representing raw values such as human and robot posture data, which are necessary for handling gesture input and controlling robots. To address this issue, we propose *Picode*, a text-based development environment augmented with inline visual representations: photos of human and robots. With *Picode*, the user first takes a photo to bind it to posture data. She then drag-and-drops the photo into the code editor, where it is displayed as an inline image. A preliminary user study revealed positive effects of taking photos on the programming experience.

Author Keywords
Development Environment; Inline Photo; Posture Data.

ACM Classification Keywords
H.5.2. Information interfaces and presentation (e.g., HCI): User Interfaces – GUI; D.2.6. Software Engineering: Programming Environments – Integrated environments.

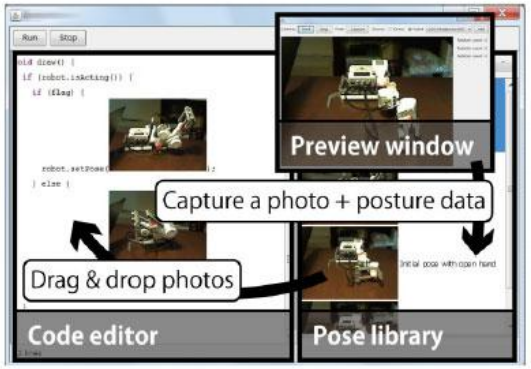
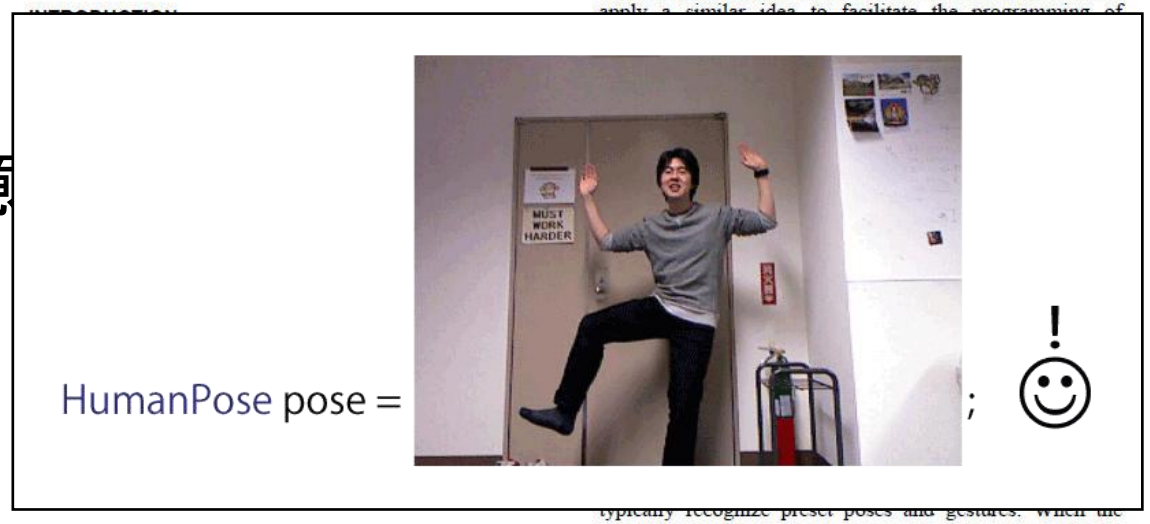


Figure 1. Overview of *Picode*

method by including an image that shows a concrete example of what the method does. Yeh et al. presented a development environment named “Sikuli,” with which the programmer can take a screenshot of a GUI element and paste the image into a text editor [12]. In Sikuli, the image serves as an argument of the API functions. Our goal was to



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typically recognize preset poses and gestures. When the programmer wants to recognize her own poses and gestures, however, she has to record the examples outside the development environment. On the other hand, our development environment is designed to support the entire

デモ動画

- 実は査読結果に**重大な影響**

[Kato, et al., 2012]

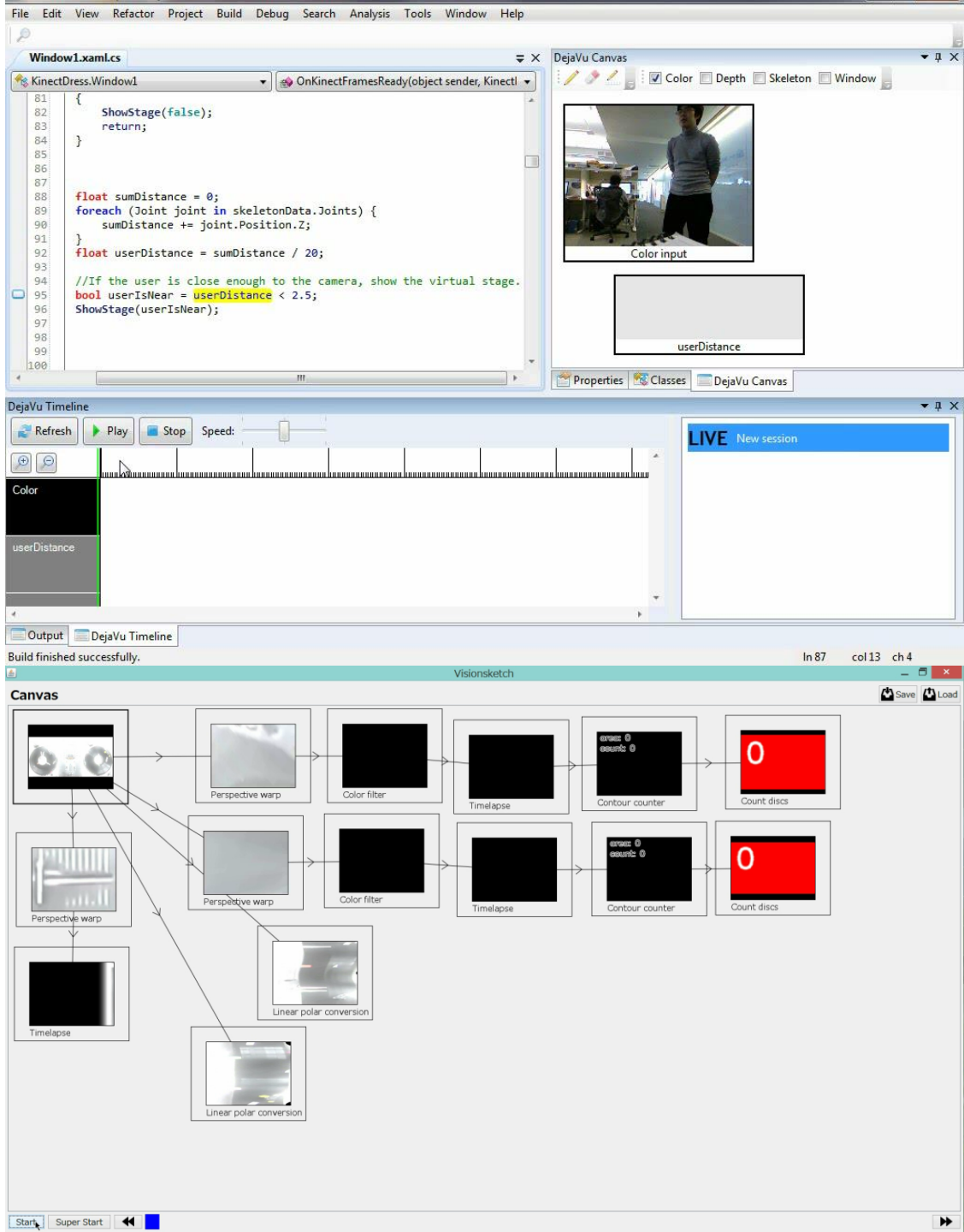
Implemented well enough?

本当にシステムを作ったのか？

Deployed well enough?

本当にまともに動いたのか？

[Kato, et al., 2014]



論文+動画

- コンテクストを保持したデモが可能



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Figure 1. A dynamic illustration authored with Draco, capturing the living qualities of a moment with continuous dynamic phenomena, yet exhibiting the unique timeless nature of a still picture. This animated figure is best viewed in Adobe Reader.

ABSTRACT

We present Draco, a sketch-based interface that allows artists and casual users alike to add a rich set of animation effects to their drawings, seemingly bringing illustrations to life. While previous systems have introduced sketch-based animations for individual objects, our contribution is a unified framework of motion controls that allows users to seamlessly add coordinated motions to object collections. We propose a framework built around *kinetic textures*, which provide continuous animation effects while preserving the unique timeless nature of still illustrations. This enables many dynamic effects difficult or not possible with previous sketch-based tools, such as a school of fish swimming, tree leaves blowing in the wind, or water rippling in a pond. We describe our implementation and illustrate the repertoire of animation effects it supports. A user study with professional animators and casual users demonstrates the variety of animations, applications and creative possibilities our tool provides.

Author Keywords

Sketching; animation; kinetic textures; direct manipulation.

ACM Classification Keywords

H.5.2. [Information interfaces and presentation]: User Interfaces - Graphical user interfaces.

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INTRODUCTION

For centuries, people have attempted to capture the living qualities of surrounding phenomena in drawings. Sketching, in particular, is a popular art medium that has also been widely adopted as a powerful tool for communication, visual thinking and rapid design, due to its minimalistic yet greatly expressive nature [6]. While sketches do afford many techniques to convey dynamic motion of objects, such as speed lines [20], arrows [14] or afterglow effect [5], they are inherently static. The goal of this paper is to enable artists and casual users alike to enrich static illustrations with intricate and continuous animation effects, while preserving the unique timeless nature of still illustrations (Figure 1).

In recent years, researchers have developed new tools and techniques for casual animation authoring using sketching [11, 21, 33] and direct manipulation [15, 29]. Such tools typically support basic animations, where motions are defined for individual objects, and then coordinated using a global timeline. In contrast, many natural phenomena are characterized by the coordinated motion of large collections of similar elements, like snowflakes falling to the ground, water drops dripping out of a fountain, or school of swimming fish (Figure 2). Animating large collections of objects with flexible control is still tedious and cumbersome with existing sketch-based animation tools.

For authoring the animations of object collections, complex software and workflows are often required. Graphic researchers have developed content-specific tools [26] and models [22, 28] for particular phenomena, but these models are highly specialized and geared towards physical accuracy for professional animators. Furthermore, defining

[Kazi, et al, 2014]

文字表現

抽象的ロジックの精密な表現が得意

画像表現

具体例を分かりやすく表現できる

動画表現

ものごとの流れや時間変化を分かりやすく表せる

知の構造化

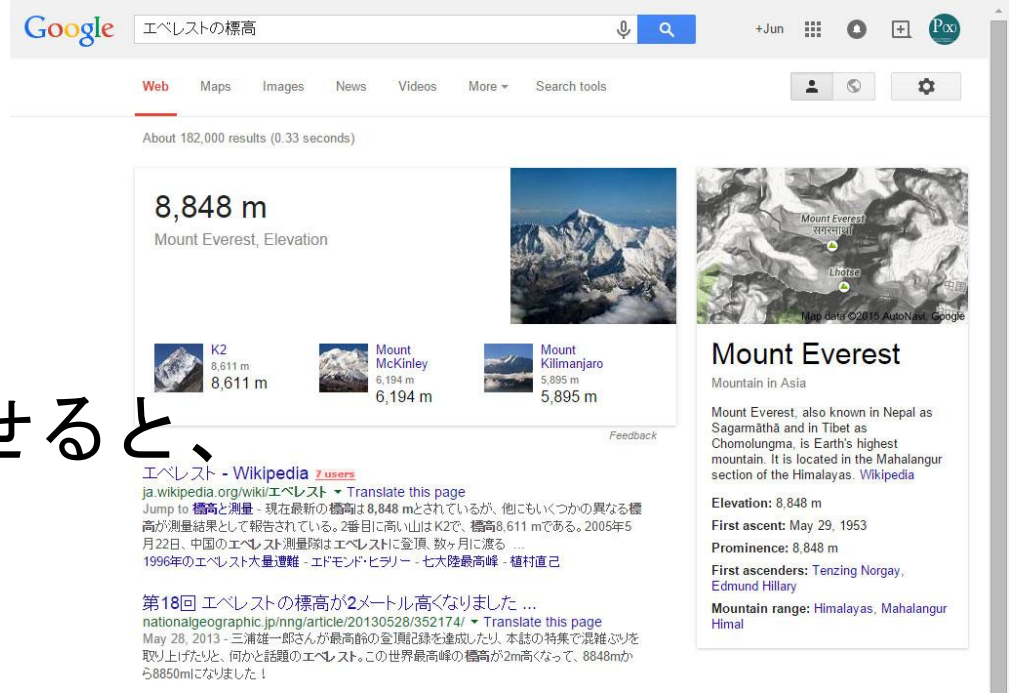
- 東京大学 小宮山宏 元総長が提唱

“構造化とは、**要素と要素の関係性**を明らかにすることである。関係性には、階層性、因果性、関連性、類似性など様々な種類がある。知の構造化の目的に応じて相応しい関係性に着目し、分散する膨大な知識を関係付け、**知識システム**を構築することを知の構造化と呼ぶ。”

[http://www.cks.u-tokyo.ac.jp/]

→知識システムの構築には至っていない

検索



- データベースにキーワードを問い合わせると、関連情報が得られる
- Google **Knowledge** Graph

> 1. 経験または教育を通して人が獲得した専門的技能。ある主題についての理論的または実用的な理解。
> 3. 事実または状況を経験することで得られた認識または知悉。
>
> [Wikipedia]

→Googleが知識を有し、人は必ずしも知識を得られない

これからの科学

- 人間にインタラクティブな体験をさせると、
  人間の中において知識が獲得される
- 科学が普遍的な知識の総称であるならば、
  それはインタラクティブシステムとして記録され、
  伝承されるべきである
- 必ずしもパテントなど既存の仕組みとはコンフリクトしない

→計算機の助けを借りた効率的な知識の獲得を目指すべ

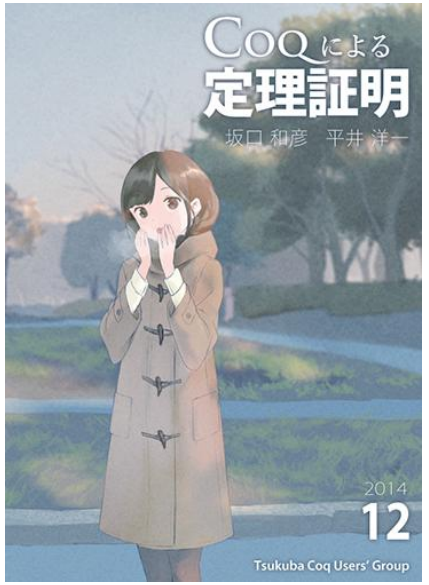
結果の共有からプロセスの共有へ

- すでにコンピュータ科学、生物情報科学の周辺分野などでプロセスの共有に向かう流れができつつある（事例募集中）

情報科学・数学

証明支援システム Coq





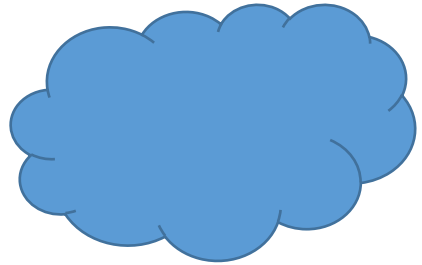
生物学（ハードウェア

オープンソース生物学ツール DIYbio

生物学（ソフトウェア

「数字を見せろ」から「コードを見せろ」へ 可視化ツールCytoscape

論文+システム: Science as a Service



- システムの実体はクラウド上にある
  （すべて開示する必要はない）
- 読者はシステムのインタフェースに触れながら、理解を深めることができる

→ 科学の透明性が向上する
研究成果の再利用性が増す
執筆＝プログラミングにな

アルゴリズム

ソースコード・実験プロトコ

データ

実験過程で得られる情報

インタフェース

アルゴリズムを試したり、
データを閲覧できる



統合開発環境

- 今後、より多くの人々が「プログラミング」を行うようになる
- より広義にプログラミングができる環境が重要である

- 加藤の研究: http://junkato.jp/ja
- Unity (ゲーム≒インタラクティブシステム)
  - Unity Solution Conference
  - おうちハック
- Microsoft Visual Studio (Compiler as a Service etc.)